

"Accounting for Multiscale and Conscious Psychobiological Phenomena via Quantum Entanglement"

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all right welcome back everyone my name is Devon Bay um I'll be presenting uh accounting for multiscale and conscious psychobiological phenomena in the active inference Paradigm so for an overview I'm going to be going over the Markov blanket trick and then critiques of the Markov blanket and address how some of the previous work of Chris fields and Friston partially addresses these critiques um before finally going over uh problems of consciousness new approaches and theories on Consciousness and how these fit within the Fe and active inference and then provide some ways to update the framework uh based on this information I'm also going to uh take an epistemological perspective on modeling and I'll focus on how both the conceptual mathematical formalizations of superposition and entanglement can maybe alleviate some of the problems that are identified in the critiques so um the Markov blanket since we are on uh limited time here um I will assume that most of the people here are already familiar somewhat with the Markov blanket or active inference framework um but I'll give a brief overview of the Markov blanket focusing on some of its mathematical Frameworks as well as some of the Assumption assumptions implicit within the the framework so um for the mathematical overview given a state s of a model it's Markov blanket is the set of sufficient States within the network needed to infer that state and it essentially partitions autonomous and external States and uh is helpful in identifying which variables are conditionally independent from each other and expresses The Joint probability between external and internal States um as mediated by blanket States so the assumptions here is that there sparse coupling between the different states in the Markov blanket uh in yeah in in the uh in the overall system and that there is uh somewhat of a delineation between external and Internal States and uh that these external internal States can be properly partitioned via the Markov blanket so uh the implications here the Markov blanket is metaphysically distinct from Pearl blankets uh at least in the context of active inference or for energy principle I think this was uh raised by um uh by

uh boomerang or and and his colleagues um in the uh in the Markov uh in the emperor's new Markov blanket paper um and so this essentially means that they are um they are uh they differentiate themselves from Pearl blankets by nature of their application and the frisen blanket itself by nature uh takes on assumptions like the existence of external internal sensory and active States um and so this is the the main sort of differentiation also uh the frisen blanket is not providing a metaphysics um at least uh at least in my interpretation and so uh so we assume here the existence of distinct states which may be modeled um and this means that there are distinct States which are properly separable and that there are either epistemological or ontological grounds for doing so and so the main problem here I think is that there is not enough justification to argue that the the Markov blanket partitioning is an epistemological or um you know in in a more broader sense an ontological um boundary and that is uh you know causally um correct or it is it causally captures is able to causally capture causal relationships between variables um because the existence of sensory um of sensory active in internal States um and external States is something which can be disputed and which I'll discuss later on in this talk and so uh another implication of the Markov blanket is that it allows the modeling of recursive structures and Dynamics by use of the re-normalization group for example which um apparently allows the Markov blanket to capture multiscale phenomenon um and finally the question of whether the Markov blanket adheres to the Bayesian Occam's razor um it would simply depend on the choice of parameters for the specific model and does not inherently implicate the Markov blanket so um critiques of the Markov blanket like I mentioned earlier the ontological critique of the Markov blanket is leveraged by both Brunberg and Raja in their respective papers although as we've seen in responses to these papers frankly depends on the ontological interpretation of active inference and uh the free energy principle and if as Kerstein in his response to Brunberg's paper claims the Markov blanket should not be interpreted literally then this problem partially goes away however the epistemological status of the Markov blanket is still under question as raised in a part of Raja's paper in so far as the explanatory power of the models under some pressure so chiefly there's a broader philosophical question about the purpose of models and this is not necessarily unique to FP or active inference or the Markov blanket but nevertheless since it affects all models these are issues are raised um are raised in in in uh in in the context of the Markov blanket as well uh next we also see that there is a mathematical uh problem with the selection of Markov blanket States um because we see that the uh for example uh Dynamic causal modeling in Neuroscience does not demonstrate that uh that models provide scientists with a plausible

account of the behavior of real world biological systems since successful applications of causal modeling does not give us any reason to believe that neural systems plausibly for example minimize free energy or that they are active inference systems and so uh this means that even though the model may be applied successfully and may predict some uh Behavior successfully it doesn't necessarily mean that the model is number one an accurate representation of what's ongoing with uh you know the internal dynamics of the system and number two that it would therefore be able to capture accurately all of the internal workings of this of the system um and so uh so yes and on this mathematical side as well we also see that uran Elon published in 2023 um a paper on the constrain disorder principle and how that would uh possi dictate neuronal function and so Elon points out that despite the stochasticity exhibited by the neur nervous systems operation there's not necessarily a high level of control since Randomness at the lower levels never stops and thus it dictates a paradigm of reaction not action by organisms um he uses the free energy principle but does not use it to justify a form of active inference which would be minimizing certainty but rather argues that the constraint disorder principle um uh is is more preferable where a process of balancing disorder and order is more inherent to systems operation in highly Dynamic random environments so whether all environments are highly Dynamic and random and at what scales this Randomness can be asymptotically ignored becomes uh or becomes irrelevant is still for question however the fact that these boundaries are not clear and in fact that they arguably cannot be made clear for example via the um existence of Maxwell's demon or limits on computability Etc should point us in the direction of choosing better models and better parameters for these models so what can this tell us about uh the free energy principle in the marov blanket first it points to broader problems with the free energy principle but these still carry over um implications uh the sorry these still carry implications for the ways in which the free energy principle epistemologically treats the marov blanket mainly to delineate uh a cross in between scales and uh causal uh relationships in the later portion of this talk I'll explain why entanglement and superposition are suitable replacements for the marov blanket when they actually offer a greater flexibility without constraining the mathematical and philosophical operations and notably Chris Field's paper on the FP for generic Quantum systems makes implicit use of the same some of the same affordances offered by superposition and entanglement but takes a very specific interpretation of quantum systems and I think limits the flexibility of of such modeling tools by adhering to the holographic principle um so uh now onto the uh epistemological uh status of models which I think is important to raise here because the

because the question of whether the marov blanket can accurately Encompass multi-directional uh what I'll call horizontal Pathways vertical and horizontal Pathways even without the presence of clear attractor States is still under question as is the more General Assumption of ergodicity within the Fe framework and so um as experimental evidence from physics shows um systems in general not simply at macro or micro states are afflicted by complex Dynamics such as replica symmetry breaking which fall into a sort of gray area between ergodicity and non-ergodicity and thus render the determination of of ergodicity versus non-ergodicity much more complicated even with suggestions such as local ergodicity as outlined by ramstead in uh in the 2018 article so um I think it's important to outline the epist ological status of models because um it's important for clarifying whether the marov blanket itself is problematic within the uh the FP framework and whether it can potentially be reworked so from a beijan perspective models should want to minimize both epistemic and alator uncertainty where epistemic uncertainty is the reducible part of the total uncertainty and alator uncertainty is basically the irreducible part active inference models therefore should minimize uh both epistemic and alator uncertainty um and represent possible model configurations and parameters via prior hypothesis space where the choice of model is informed both by where we believe the qual truth is located or at least the uh the most truthful uh model is located as well as the model constraints so here we see um in green is the ideal situation where we have an ideal selection of model with some with an ideal convergence to some truth X um and then uh the purple shows an example here where uh our prior hypothesis space does not accurately Encompass the true model and hence by extension the true model is not included in the posterior finally the pink shows where the posterior does not converge to the true model um and so this is illustrating how problems can arise in these two areas first the hypothesis base does not properly Encompass the true model X which is uh the purple or blue example and then second the posterior space does not properly converge to the true model X which is our pink uh Pink example here and so constraints on X uh depend on if you're a realist or an anti-realist so for example if you believe that there is some ground truth that is universal and um unchanging and it's you know physically uh accessible then of course the true model would be a a real true model but uh if you're more of an anti-realist then it would depend on uh you know your your hypothesis space and you know of course whether you believe that troops can be can be accurately represented or even be accessed um and that they're Universal or whatever so yeah is now now the question is is the marov blanket the proper choice of parameter for the free energy principle and active inference so the purpose of the marov

blanket is to provide a statistical boundary which delineates a system from its environment and ensures that variables are conditionally independent of each other. That certain variables are conditionally independent of other variables in the network and it formalizes the belief of the prior that the system and its environment are separate either ontologically or epistemologically. Physicists like literally or even causally right so Umberto Colombo and Palasio actually indict the Markov blanket and the active inference model by saying that it achieves maximal generality for minimal biological plausibility because these boundaries can be drawn sort of arbitrarily and even the existence of cellular membranes for example does not necessarily constrain causal relations between or across scales so notably uh Noble argues that organisms are open systems that operate far from equilibrium and so again drawing from Noble's arguments uh Elon in the 2023 paper argues that organisms develop numerous emergent boundaries between levels which are not present at the single cell level so even though they may have distinct functional purposes boundaries like the skin intestinal tract or even vastus medialis depend on exchanging matter and energy through cellular membranes and so boundary is thus Dynamic interactive not passive and cannot always be defined anatomically so for example electric gradients and ion concentration gradients are actively generated across cell membranes where interactions across boundaries determine an organism's functions uh not just parts of not just the composition of the boundary or uh or the genome so of course one way this has been explored in the free energy principle and active inference is via uh the sort of recursive Markov blanket but uh we see that this sort of recursive modeling is not able to properly capture multiscale relationships and interactions across boundaries where essentially we would have uh you know Markov blankets that are that are causally implicated in more than one uh in more than one way that that can vary in more than one way that can be described by simply the repertoire of active external and internal states uh and sensory states and so this is sort of where the case for entanglement comes into play because uh entanglement is uh it has been argued for as a way of causally representing relationships between variables in uh in biology so first of all we see that organizational boundaries of living systems are open and flexible such that these boundaries need not be co-extensive with the organism's bodily boundaries number one so again I guess this is just to drive home the point uh that the Markov blanket should not be taken literally um but we also see that biology is inherently Quantum so uh so these um so these uh these processes of uh cellular differentiation for example uh are under are fundamentally underlied by or sorry under fundamentally you know determined by or perhaps uh sort of controlled by

uh Quantum processes that dictate you know um what chemical bonds the cell is able to break for example or how certain concentrations of ions for example flow across um across the cellular membrane and the permeability of the membrane as well can also exhibit Quantum effects so as you know the field of quantum biology grows I think we will increasingly see that the quantum levels is extremely important for dictating how uh organisms evolve and especially if you know um Quantum Consciousness theories continue to gain support uh we could see that uh Quantum processes may also be implicated in Consciousness and so causal entanglement addresses a sort of asymmetry um between what were previously considered linear processes um and um the sort of asymmetry across in between scales so uh I basically argue that causal entanglement is a counterintuitive form of aam's Razor because when we have entangled variables we don't need to always explicitly represent the dependencies of these variables um with each other so um uh that means that we are we can you know in an infinite dimensional Hilbert space or a finite dimensional Hilbert space represent the variables uh in a way that does not necessarily dictate uh specific boundaries or um or constraints on these variables which may impede the ca of modeling of uh certain phenomena so as the scale and complexity of the model increases representing relationship via entanglement becomes much more appealing and of course uh you know Quantum Computing also offers ways to um to speed up um uh the the computation of machine learning models like act like an active inference so uh vti actually puts forward a case for causal entanglement using uh DNA and essentially we see that uh the in lement can serve as a form of um uh a form of of sort of outlining the the total the sorry the different causal dependencies in a system without necessarily explicating them without putting too much emphasis on the uh the specific properties of the causal relationships if we know that causal certain things are causally implement or sorry are causally implicated then we can take the sort of rough approach of assuming that they might be uh you know entangled to some degree um and so basically entanglement can serve as a method for representing causal Relationships by conflating correlation with causation and so the um the in the paper vei uses chromatin structures during the interface period of mitosis to show how and basically show how it can be characterized in terms of interactions between entities at uh three different levels of biological organization so essentially the the parts or angles between the linkers the uh in the chromatin and um then uh the the chrom structure or the topological invariant structure and then finally the the in the feedback loop between how the whole which is the the overall chromatin structure depends on context properties in the feedback loop uh for example via histone dilation or histone

methylation and so essentially uh we see that they argue that DNA is a total both a total and direct cause when being manipulated where it is a total cause because DNA is implicated in its in its own manipulation and it's also a direct cause because um it's uh it's an intervention on DNA will change uh will change the the you know this the functional structural properties of the DNA um with with without changing necessarily L um additional uh non lo non-local you know properties of whatever substrate surrounds the DNA and so specificity still allows linear models to capture causal relationships in terms of you know isolating specific variables but when we increase the scale and complexity of these uh of the model for example or what we include in the model it becomes increasingly more difficult to separate out and to sort of distinguish okay where exactly does a specificity lie and so uh this takes me to this differentiation between repertoire and connectedness when it comes to causal specificity so repertoire is basically how many possibilities exist on each side the cause side and the effect side and we can essentially in in a modeling context take this to mean the number of different values that an effect variable and a cause variable can take meanwhile the connectedness is how many or how the possibilities on both sides are connected in particular relevant ways so we can see uh again in the modeling C test that connected connectedness is a condition for a given function connecting the values of cause and effect variables and so repertoire does not entail connectedness and connectedness does not entail repertoire which is important to note um because essentially we see that um that uh repertoire and but by splitting causal specificity into these two independent properties we may not be able to compare causal relations from a single point of view and therefore uh you know causal entanglement will may allow for the comparison of both without necessarily giving up you know one one for the other and uh and of course uh Ru notes that you need both repertoire and connectedness in order to come to a complete causal understanding of you know whatever processes are implicated or you're examining within the system and so the the form of entanglement between uh you know repertoire and connectedness um would would allow you to express uh both of them in the system simultaneously and potentially uh you know um extract results from uh from either of them and so uh what are the practical constraints and limitations on this uh this approach well first of all we should uh view we might view it as a case similar to Quantum versus classical Computing where as the complexity class of the problem increases Quantum Computing becomes more appealing so utilizing entanglement may not be computationally feasible right now given Hardware constraints and um modeling using the entanglement May aisc uh the extraction of direct cause of relationships but um as Noble has pointed out and this previous

researchers have pointed out the relationships is a very difficult task and epistemologically it's not clear whether this is even possible once the system surpasses a uh given uh given you know point of complexity where it's simply where the dependencies between uh and across scales simply become too large to um to properly uh identify you know direct cause or relationships and so um essentially uh this this entanglement approach may also become computationally infeasible when uh we have lots of uh you know dense variables that are not sparsely coupled but I think because uh you know the Markov blanket also you know inherently relies on the sparse coupling and sparse coupling in neuronal Dynamics has been shown uh entanglement can be taken as an as a sort of replacement for the Markov blanket um and so I'll just uh get into some previous work involving the free energy principle in quantum mechanics so Fields uh Chris Fields integrated Ed Quantum information Theory with the free energy principle and proposed that physical interaction is a form of information exchange uh where the holographic principle can essentially be used to formalize boundaries between systems uh additionally Warier investigated the potential for tensor networks to represent um generative active inference models so tensor networks of course can offer computational benefits When run on quantum computers and the tensor Network itself was originally um designed to be used uh in um in quantum mechanics um so specifically for Fields approach uh with the quantum reference frames we see it's a different way of approaching the Markov blanket um which still formalizes boundaries using the holographic principle so it's still assuming that we can properly distinguish the boundaries between uh you know organisms and their environments and that these boundaries are both causally uh necessary right so that there's some causal impact at the boundary uh that cannot be ignored and that the boundary itself uh is you know properly distinguishable um so now I'll just I'll go into uh some theories of sorry the overview of consciousness and how it's implicated in uh both Fe and active inference and how it fits into the proposed uh framework for um for for uh for utilizing entanglement so uh the minimal causal mechanisms of Consciousness we see that there are regions in the brain that are implicated in Consciousness where you know if there uh if these regions of the brain are damaged or not functioning properly this leads to changes in the overall phenomenal and conscious uh perception of individuals so we also see that non-equilibrium macroscopic brain Dynamics are a robust signature of Consciousness so at some level Consciousness uh can you know statistically and um uh and physically be proven to be the sort of higher order uh um process or or dynamic that is going on in the brain and so uh Consciousness and the Markov blanket so how does Consciousness interact with the Markov

blanket and the free energy principle does the link between phenomenal perception judgment awareness observation Etc disrupt the epistemological application of the Markov blanket and do competing theories of Consciousness pose problems for active inference so uh on the first question I think that when it comes to the Markov blanket and free energy principle Consciousness presents you know as I guess with all neuroscientific theories uh is the sort of uh throw a wrench in things when we are trying to uh uh consider how you know conscious States or different levels of conscious States may not be easily or neatly put into boxes of either internal or sensory States because again you're assuming that the organism has either the organism itself has some capacity to differentiate between these states and that the states themselves can be placed into this sort of uh into these these sort of categories for representation by the Markov blanket so I think this is especially apparent when we are discussing phenomenal perception judgment awareness uh observation Etc because um these types of phenomena are very very broad ranging in terms of their uh phenomenal activity and their corresponding correlation sorry their correspondence to underlying neuronal Dynamics um and of course the research there is still ongoing but uh based on you know whatever what research there is into uh awareness for example attention or observation um there there there seems to be um varying sort of uh degrees of complexity and uh also degrees of causality when it comes to you know which areas of the brain are implicated in minimal self-awareness and to what extent these boundaries uh initially observed can be modified um or you know modified either internally by the person The Observer Modified by some external uh external causal Force for example uh like anesthetics and so the competing theories of Consciousness sorry do competing theories of Consciousness pose problems for active inference and I would say depends on the mechanism of Consciousness right so it depends on how these theories lay out Consciousness because they may or may not implicate active inference for example uh theories which are more volitional on the on the more volitional side that attribute um you know Consciousness to volitional activity which in itself is uh maybe distinct from you know uh free energy minimizing um or uncertainty minimizing activities then we would see that uh that yes these competing theories of Consciousness do or would pose problems for uh for active inference and another Dynamic that I want to explore here is uh the unconscious right so experimental evidence supports a sort of minimal unconscious where we can broadly distinguish between type one and type two thinking so type one thinking is unconscious processing and type two thinking is conscious processing and um researchers have sort of found that there is a form of dynamic unconscious which sits between the conscious

processes that are you know directly under observation or awareness and then those perhaps autonomic processes for example where uh they're they're entirely uh unconscious and so the dynamic unconscious Theory essentially posits that the uh The Observer or the the conscious agent themselves has some capacity to uh to guide their awareness to processes that were previously um previously unconscious so uh you can essentially make previously unconscious processes conscious and again the implications for the mark of blankets uh neat delineation between sensory active and internal States um I would say this this sort of frustrates uh the the um the thinking behind Okay the thing behind trying to separate out uh the uh these um these these different states right and so theories of the brain should incorporate a transition from unconscious to conscious awareness um and of course this is extremely difficult because we don't have really a solid understanding of what would what Consciousness even is what are the minimal phenomenal sort of uh requisites of Consciousness and how sort of expansive the operations of the unconscious are and whether those unconscious operations are therefore implicated in conscious processing as well for example with uh underlying cognitive biases so uh what are the implications for the free energy principle so both conscious and unconscious processes should be modeled however the distinction between uh sensory and internal states does not do uh this uh these conscious and unconscious distinction I think it doesn't do adjust so why is that the case well first conscious processes are sometimes unconscious processes are processes that may inherently be uh be sort of semiconscious for example when someone is uh is scratching their their leg for example right the the distinction between uh conscious and unconscious processes is one that is fundamentally dynamic and therefore um we don't know whether there even is a boundary between the awareness of scratching leg and and not scratching leg and especially if that boundary is uh is sort of um is highly Dynamic for example uh when someone is slipping in and out of Consciousness post anesthesia um and so we see that unconscious and conscious processes could influence active inference in ways that are are particularly unique to them so for example unconscious processes uh could be represented as uh as you know as priors within the model um but that doesn't always necessarily have to be the case because these unconscious processes could also be interacting as um as as internal States for example or they could be uh they could be present even as active States once they are made uh made conscious um and so the mechanisms of Consciousness uh various uh theories which have been proposed could provide a mathematical basis for expanding active inference and the free energy Principle as well so I think this is very important to note because uh theories of Consciousness um sorry novel theories of

Consciousness that incorporate you know more Quantum information side or that correspond to um you know the electrical sorry the the electrical um the electrical correlates of Consciousness for example um are extremely important for uh for sorry May provide mathematical Frameworks for updating um active inference or you know potentially May provide a useful uh mechanisms um for modeling specific conscious processes that could then be incorporated into uh the framework of inference um and so see and so yeah so uh I think I'm missing a few slides here but uh basically the the the overall theories and most promising theories of Consciousness um uh that that are that are upcoming are you know the sort of um the the electrophysiological and um Electrical uh the electrical field uh sort of theories of Consciousness which posit that Consciousness is actually a result of the electrical Dynamics in the brain which of course uh you know would need to be a case as as for for any minimally causal um sort of uh representation of Consciousness but the presence of electrical fields and the implication of these fields in Consciousness is uh you know a particularly new Avenue of research and we see that for example firing patterns in the brain have produced uh quasi Crystal representation and and can be represented as well um in in in the Taurus uh for example so these uh higher dimensional uh uh Dynamics within the brain correlating to Consciousness or um or these sort of unconscious processes will be extremely important in um in determining you know what processes are relevant for Consciousness and how these might be implicated in uh in for example or uh thought patterns you know traversing whatever sort of dynamic landscape that exists in the brain which again may be determined by these higher order or uh these higher magnitude or dimensional structures so um overall we see that investigating the mechanisms of Consciousness I think is fundamentally necessary for uh for FN active inference because these uh structure um or have the potential D structure uh uh thought and the ways in which uh certain uh you know energy landscapes for example in the brain are uh are traversed or are um are fundamentally interact with each other and so uh the areas for future work here um would be to unpack the philosophical dimensions of active inference which would include implicit assumptions about intelligence sapience and sentience so for example you could utilize Denny Jean's conception of intelligence which um which argues for a a a a uh sorry a relationship between the model and the modeling so and sorry the model and the modeler um as a as a direct form of intelligence there or you could utilize Imon Don's theory of individuation as a more productive method for explaining uh the boundaries um and cellular differentiation which I've critiqued previously uh you could also utilize Quantum Computing to take advantage of quantum specific algorithms and uh to investigate

uh Quantum biology um um with an increasing uh with increasing scrutiny example and and discover ways in which Quantum biology is implicated uh in you know processes of Consciousness active inference and in the brain and so essentially the the the overall Theory here when it comes to when it comes to uh oh when it comes to um to to active inference and uh and uh multiscale phenomena and uh and Consciousness is that we can essentially um if we take that the boundaries between the the system and these systems are not actually uh ontological or physically causal causally implicated um and we utilize this sort of uh causal entanglement to represent relationships between uh you know different parts of the system uh Consciousness can be taken as a form of observation along uh specific uh you know um causal trajectories within the system that would essentially collapse the superposition of these entangled states which would produce you know specific uh outcomes and so that's a very broad sort of method for integrating um Consciousness with you know Quantum phenomenon and entanglement but I think that there are many uh many many other sorry different there are many there are different methods of integrating quantum mechanics with uh with the Paradigm of active inference for example that don't just that are that are distinct from the holographic principle and do not rely on the explicit representation and encoding of information in uh in boundaries for example um and when it comes to for example these unconscious processes as I've outlined in the abstract these unconscious processes could be represented more so by you know the minimization of uncertainty um uh via you know the free energy Principle as always ongoing processes similar to elon's argument whereas conscious and more volitional um uh actions would be represented by the sort of observation on this Quantum system Where We Are uh where we are collapsing the superposition of specific entangled States and so I think that in this in in that way entanglement offers two sort of uh sort of benefits to the active inference Paradigm the first is that it allows for some flexibility when we are talking about uh different variables and causal connections between and across scales because we can entangle two variables that are causally distinct um sorry that are that are uh scale that are that are distinct across scales for example but which may be implicated in some uh feedback loop for example um and this entanglement can then uh represent the causal dependencies between these variables on the other hand um if Consciousness you know is a sort of quantum process then it would allow allow uh it would allow for the integration of whatever Quantum processes that are ongoing in in whatever you know Quantum Consciousness uh um so if if you believe that Consciousness collapses the wave function for example then uh then incorporating entanglement would be a mathematically feasible um technique for

incorporating this uh this form of valuation where essentially the uh the conscious agent is collapsing a specific wave function and that leads to uh you know that allows them to perform uh active inference for example by minimizing uh specific uh probability densities so um yeah that's overall uh the the scope of my talk today um and yeah I hope uh everyone uh enjoyed that and uh if you have any question questions please let me know thank you Devin great times so I'll ask one question you'll give the closing word for the live streams for the whole Symposium so you're the only undergrad presenter what would you say excited you to get into the game and what do you say to younger Learners to come um um I am mainly interested in uh um the sort of mathematical and philosophical um uh I guess um ideas behind uh active inference and um I was I'm particularly um uh happy to see that the active inference ecosystem has you know thriving uh philosophical and uh and mathematical um Community um so when it comes to uh um younger uh um uh participants who may be interested in active inference I would say uh you know keep uh pursuing um what you're interested in and um if you have any particular areas of research or ideas you should always um be be looking to share them I think and uh be looking to expand your horizons so yeah cool well looking forward to hearing more from you and from others so thanks again Devon for bringing so much together in this last Capstone talk thank you so much for inviting me to uh give the talk and apologies uh I was a little bit short on time to uh to put together the slides so uh this was a lot uh messier than I would have liked unfortunately but uh yeah real life is like that all right thanks Devon peace thank you so much